This military standard is approved for the Department of Defense and is mandatory for use by all Departments & agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. NAVY-AS

Other Cust
USAF - 11
ARMY - EL

Procurement Specification
MIL-C-81703

International Interest

SUPERSEDES:

TITLE
CONNECTOR, RECEPTACLE, ELECTRIC, SOLDER TYPE, WALL MTG. FLANGE, PUSH-PULL COUPLING, SERIES 1 (CLASS E, P AND J)

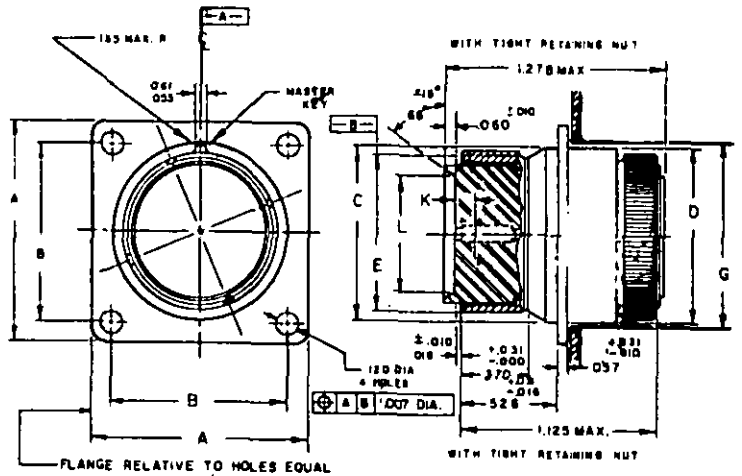
MILITARY STANDARD

MS 3130

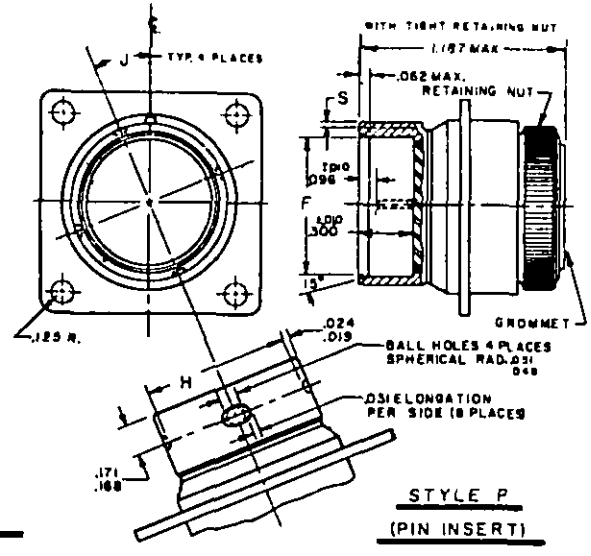
SHEET 1 OF 3

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5935-1761-1



STYLE S
(SOCKET INSERT)



STYLE P
(PIN INSERT)

CLASS E

ENTIRE STANDARD REVISED.

SHELL SIZE DASH NO.	A MAX. LENGTH SIDE	B BSC MTG. CENTERS	C MAX. DIA. FRONT	D MAX. DIA. REAR	E +.006 -.004 SHELL O.D.	F +.003 -.007 SHELL I.D.	G MIN. DIA MTG. HOLE	H +.007 -.008 BALL HOLE SPACING	J BALL HOLE LOCATION	K POINT OF SPRING CONTACT	L INSERT SEAL I.D.	S +.003 -.004 HEIGHT OF KEY	FOR INSERT ARR. SEE DWG.
- 3	.898	.625	.578	.625	.435	.364	.632	.398	30°	.136 MAX MEASURE WITH SQUARE NOSE PIN HAVING BASIC DIAMETER OF MATING CONTACT	.244	.045	MS 33698
- 7	1.021	.719	.691	.694	.870	.480	.698	.531			.370		MS 33699
- 11	1.114	.812	.828	.828	.704	.621	.835	.666			.510		MS 33700
- 19	1.208	.906	.953	.953	.843	.760	.860	.804	22½°		.650		MS 33701
- 27	1.302	.968	1.136	1.078	.998	.915	1.143	.958			.803		MS 33702
- 37	1.458	1.187	1.266	1.266	1.120	1.037	1.273	1.081			.927		MS 33703

EXAMPLE OF PART NO. MS 3130 E 19-50 S Y

MS NUMBER | CLASS | SIZE | INSERT ARRANGEMENT | CONTACT STYLE | ALTERNATE INSERT POSITION

DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCE: DECIMALS ±.016
ANGULAR TOLERANCE ±1/2°

NOTES:
1 TRUE POSITION (T.P.) TOLERANCES SPECIFIED ARE IN ACCORDANCE WITH MIL-STD 8
2 DIMENSIONS LOCATING TRUE POSITIONS ARE BASIC

THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID.

FED. SUP. CLASS 5935

APPROVED 14 MAY 59 REVISED (B) 1 NOV 1965 (C) 6 MAR 67 (D) 21 JULY 1971

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P.A. NAVY-AS

Other Cust: USAF-11 ARMY-EL

International Interest:

Procurement Specification: MIL-C-81703

SUPERSEDES:

TITLE: CONNECTOR, RECEPTACLE, ELECTRIC, SOLDER TYPE, WALL MOUNTING FLANGE, PUSH-PULL COUPLING, SERIES I (CLASS E, P AND J)

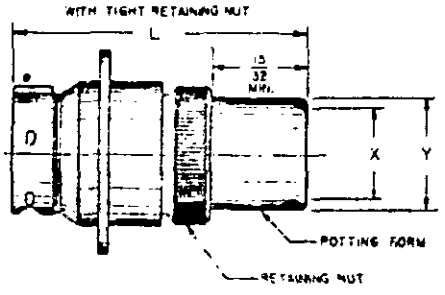
MILITARY STANDARD

MS3130

SHEET 2

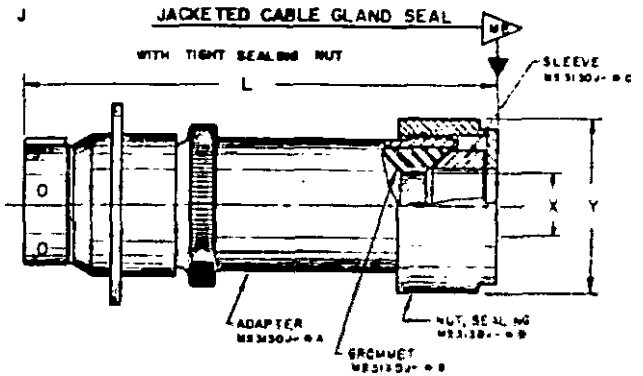
FED. SUP CLASS 5935

CLASS P — POTTED SEAL



SHELL SIZE DASH NO.	L LENGTH MAX.	X DIA MIN I.D. POTTING FORM	Y DIA MAX O.D. POTTING FORM
- 3	1.640	.261	.391
- 7		.372	.515
- 12		.482	.630
- 19		.602	.760
- 27		.725	.887
- 37		.857	1.047

CLASS J — JACKETED CABLE GLAND SEAL



SHELL SIZE DASH NO.	L LENGTH MAX.	X CABLE SEALING MIN.	X CABLE SEALING MAX.	Y MAX DIA NUT	ADAPTER	GROMMET	SLEEVE	NUT SEALING
- 3	1.641	.178	.210	.643				
- 7	2.025	.308	.413	.926				
- 12	3.093	.412	.444	.988				
- 19	3.125	.522	.554	1.113				
- 27	3.125	.600	.632	1.113				
- 37	3.656	.662	.694	1.178				

DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED. TOLERANCES: DECIMALS ± .010, ANGULAR TOLERANCE ± 1/2°

NOTES:
1. FOR CLASS J ADAPTER, SEALING NUT, SLEEVE AND GROMMET SEE SHEET 3.

MAX CABLE WEIGHTS IN POUNDS (ALLOWABLE PUSH IN DIRECTION INDICATED)

SHELL SIZE	MAX WEIGHT
7	2.0
12	2.4
19	3.0
27	3.5
37	4.0

EXAMPLE OF PART NO.: MS3130 J 19-50 S Y

MS NUMBER	CLASS	SIZE	INSERT ARRANGEMENT	CONTACT STYLE	ALTERNATE INSERT POSITION

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APPROVED 14 MAY 59 REVISED (D) FOR CHANGES SEE SHEETS 1, 2 & 3.

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P.A. NAVY-AS

Other Cust USAF-11 ARMY-EL

International Interest

Procurement Specification MIL-C-81703

TITLE

CONNECTOR, RECEPTACLE, ELECTRIC, SOLDER TYPE, WALL MOUNTING FLANGE, PUSH-PULL COUPLING, SERIES I (CLASS E,P AND J)

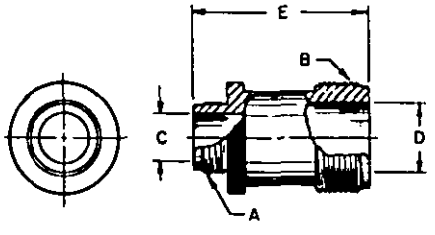
SUPERSEDES:

MILITARY STANDARD

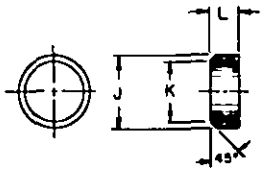
MS3130

SHEET 3

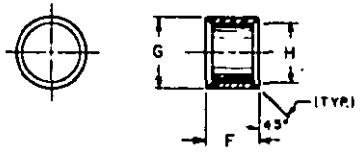
FED. SUP CLASS 5935



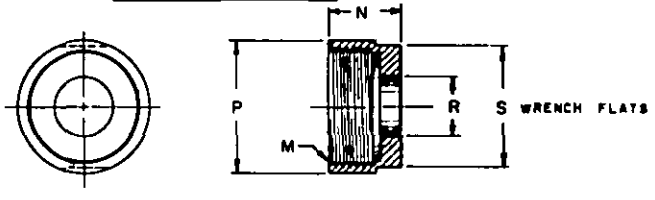
ADAPTER MS3130J-*A



SLEEVE MS3130J-*C



GROMMET MS3130J-*B



NUT, SEALING MS3130J-*D

SHELL SIZE DASH NO.	A – ADAPTER A (THD SIZE)	B (THD SIZE)	C I.D.	D I.D.	E O.A.L.	B – GROMMET F O.A.L.	G O.D.	CABLE SEAL DIA. H (MIN)	H (MAX.)	C – SLEEVE J O.D.	K I.D.	L O.A.L.	D – NUT, SEALING M (THD. SIZE)	N O.A.L.	P MAX. DIA. NUT	R I.D.	S WRENCH FLATS
-3	1/2-28 UNEF	1/2-28 UNEF	.364	.385	.719	.344	.375	.178	.210	.375	.281	.219	1/2-28 UNEF	.437	.645	.281	.563
-7	5/8-24 UNEF	5/8-20 UNEF	.495	.593	1.609		.580	.381	.413	.580	.485		5/8-20 UNEF		.926	.500	.812
-11	3/4-20 UNEF	13/16-20 UNEF	.593	.625	2.073		.672	.412	.444	.609	.500		13/16-20 UNEF		.988	.500	.875
-15	7/8-20 UNEF	1-20 UNEF	.718	.687	2.109	.437	.672	.522	.554	.672	.578	.246	1-20 UNEF	.531	1.113	.578	1.000
-27	1-20 UNEF	15/16-20 UNEF	.834	.750	2.109		.734	.600	.632	.734	.640		15/16-20 UNEF		1.113	.640	1.000
-37	1 1/16-18 UNEF	1-20 UNEF	1.037	.850	2.109		.835	.662	.694	.835	.760		1-20 UNEF	.562	1.176	.766	1.062

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED, TOLERANCE DECIMALS ±.010, ANGLES ±1/2

NOTES:
1. A INDICATES ADAPTER
   B INDICATES GROMMET
   C INDICATES SLEEVE
   D INDICATES NUT, SEALING
2. THESE PARTS ARE INTENDED FOR USE WITH CLASS J CONNECTORS.

EXAMPLE OF PART NO. MS3130 J – 7 B
MS NUMBER · CLASS · SIZE · GROMMET (SEE NOTE 1)

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